

Causal VAE-DM: Trustworthy Generative Modeling for Image Modification Based on Causal Graphs

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Abstract—Generative models like Variational Autoencoders (VAEs), Diffusion Models (DMs), and Generative adversarial Networks have shown strong capabilities in realistic image generation, but often lack interpretability and fine-grained control. To address these limitations, Causal VAE-DM is proposed as a hybrid model that combines causal inference with generative modeling to enhance transparency and controllability. By integrating the representational capability of VAEs with the high-quality image synthesis of DMs, this method enables precise editing of specific image attributes while preserving realism. Experiments on two datasets demonstrate that Causal VAE-DM outperforms existing causal generative models in both interpretability and feature-editing performance.

Index Terms—Trustworthy generative model, Image modification, Causal inference, Variational autoencoders, Diffusion models

I. INTRODUCTION

Generative models, powered by deep learning and reinforcement learning [1], [2], have gained widespread attention in various fields, like robotics, autonomous vehicles and medical treatments [3]–[5]. However, with more generative models used in critical applications, the requirement of the trustworthiness of the models becomes paramount [6]. Causal inference provides a systematic framework to enhance transparency by explicitly modeling the dependencies between variables using causal graphs [7]. While traditional causal methods rely on structured data with well-defined relationships, generative models typically work with unstructured inputs like images, where extracting causal relationships is more challenging [8].

For images, causal factors might relate to complex visual features or hidden attributes, such as lighting, texture, or object positioning, which are not easily represented by simple variables [9]. To obtain structured data from images, causal

representation learning is proposed to transform raw data, such as images, into structured variables that can be used in causal models [10]. Causal representation learning provides deeper insights into how observed features in images impact specific outcomes or decisions.

Generating high-quality images in causal generative models is challenging due to the complexity of accurately modeling causal relationships while maintaining the realism and fidelity of generated data [11]. In generative models, variational autoencoders (VAEs) could obtain representations easily but it is hard to produce high-quality images due to their smooth loss function [12]. Diffusion Models (DMs) could generate images with high quality, but it is hard to get their representations as they rely on a sequence of noise-to-signal transitions [13]. To avoid the drawbacks of these two methods, this paper proposes a Causal VAE-DM (Causal Variational Autoencoder-Diffusion Model) method to generate high-quality images with causal relationships. The integration of the Causal VAE-DM reduces dependency on extensive real-world data collection, resulting in significant cost savings. Additionally, Causal VAE-DM accelerates development by enabling the generation of high-quality, causally consistent synthetic data. The main contributions are as follows:

- 1) Synthetic data generated by Causal VAE-DM is both realistic and aligned with real-world causal relationships, enhancing interpretability and robustness.
- 2) The method generates diverse and high-quality images that capture various environmental conditions, which reduces dependence on real-world data and lowers costs.
- 3) By introducing causal factors to image generation, Causal VAE-DM improves interpretability and control, which supports transparent decision-making in applications like autonomous vehicle vision.

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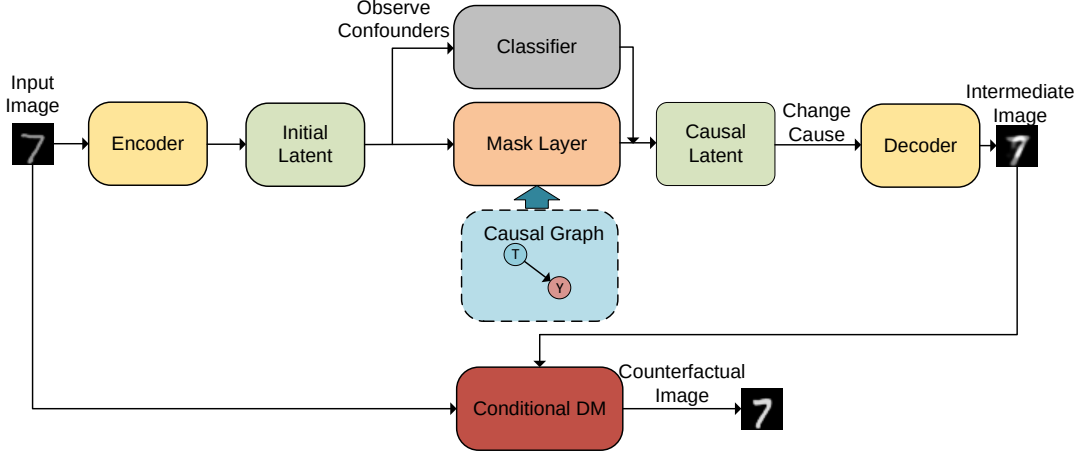


Fig. 1: Overall architecture of Causal VAE-DM

II. RELATED WORKS

Causal inference goes beyond mere correlations to uncover true cause-and-effect relationships using frameworks like do-calculus and structural causal models (SCMs), enabling counterfactual analysis through methods such as NOTEARS and LLM-based approaches [14]–[16]. This approach is particularly valuable in generative modeling, where integrating causal reasoning with disentangled representation learning enhances both interpretability and trustworthiness [17]. By incorporating causal structures into models like CausalVAE and causal GANs, the generation process becomes transparent, allowing controlled interventions and ensuring that synthetic images faithfully reflect real-world causal relationships [18], [19]. Such advancements are crucial for sensitive applications in healthcare and autonomous systems [20].

III. CAUSAL VARIATIONAL AUTOENCODERS WITH DIFFUSION MODEL

The proposed Causal VAE-DM method combines causal inference with VAE and DM to generate counterfactual images. Through the causal graph and the evaluation of the results, the changes in the generated images could be explained. In this section, the architecture, learning process and evaluation process of the Causal VAE-DM are presented.

A. Model Architecture

1) *Overall Structure*: The structure of the Causal VAE-DM is presented in Fig. 1. This diagram illustrates a structure for a causal image generative model, where an encoder processes the input image, and a causal graph represents relationships between factors. The output includes intermediate and counterfactual images generated through a causal latent decoder and a conditional DM to observe effects from specific causal changes. In this structure, the causal matrix is presented by SCM, which converts the relationship in the graph to a matrix [18]. This causal matrix $\mathbf{A}$ is introduced to encode the causal

representations. The mechanism of this process is factorized as

$$\mathbf{z} = \mathbf{A}^T \mathbf{z} + \epsilon = (\mathbf{I} - \mathbf{A}^T)^{-1} \epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I}) \quad (1)$$

where ϵ represents independent Gaussian exogenous factors, $\mathbb{T}$ denotes the transpose of a matrix, and $\mathbf{z}$ is the latent variable, which is an unobserved factor underlying observed data, processed in the mask layer.

Since the model's input consists of unstructured data, it is essential to convert the unstructured data into a structured form. In the proposed model, the latent variable $\mathbf{z}$ serves as the required structured data, encapsulating the underlying causal factors that influence the visual features in the images. A structured encoding process is employed to derive $\mathbf{z}$, allowing it to capture these causal relationships effectively. $\mathbf{z}$ is then utilized within the model to generate images that accurately reflect the causal dependencies, enabling more interpretable and reliable outputs.

2) *Confounding Factors*: Confounding factors are variables that influence both the cause and the effect within a study, leading to potential biases in estimating causal relationships. These factors create a spurious association that can obscure the true causal effect or suggest a relationship where none exists. For example, in image generation tasks, confounders such as background patterns or lighting conditions can simultaneously affect both the input features (cause) and the target outputs (effect). This dual influence makes it challenging to disentangle genuine causal relationships.

A classifier is introduced after obtaining the initial latent representation to address the presence of confounding factors, particularly if these confounding factors are easily identifiable. The mechanism of the classifier is represented as

$$\mathbf{z} = \mathbf{W} \cdot \mathbf{x}_0 + \epsilon_{\text{classifier}}, \quad (2)$$

$$\mathbf{k}_{\text{classifier}} = \mathbf{M} \cdot \mathbf{z} + \eta_{\text{classifier}}, \quad (3)$$

where $\mathbf{W}$ is the weight matrix, $\mathbf{x}_0$ is the input image, $\epsilon_{\text{classifier}}$ denotes independent Gaussian noise in the classifier, $\mathbf{k}_{\text{classifier}}$ is the output of the classifier, and $\mathbf{M}$ and $\eta_{\text{classifier}}$ are additional parameters. The classifier utilizes a fully connected network and applies cross-entropy as the loss function, effectively distinguishing between relevant features and confounding factors within the latent space that encodes abstract data features. The loss function $\mathcal{L}_{\text{CE}}$ is defined as

$$\mathcal{L}_{\text{CE}} = - \sum_{i=1}^{N_{\text{sample}}} \sum_{c=1}^C l_{i,c} \log \hat{l}_{i,c}, \quad (4)$$

where N_{sample} is the number of samples, C is the number of classes, $l_{i,c}$ is the true label, and $\hat{l}_{i,c}$ is the predicted probability that sample i belongs to class c .

By doing so, the classifier helps isolate the true causal variables, ensuring that the model focuses on meaningful causal relationships rather than spurious associations introduced by confounders. This step enhances the robustness and interpretability of the model's outputs, ultimately supporting more accurate and trustworthy image generation.

3) *Conditional Diffusion Model*: The conditional DM iteratively refines noisy inputs into high-quality images using the VAE-generated intermediate image as a guide. It denoises images step-by-step, aligning outputs with $\mathbf{z}$, and uses causal factors to control specific image attributes, ensuring real-world causal consistency. Normally, the processes of the DM are divided into the forward process:

$$q(\mathbf{x}_{1:T}|\mathbf{x}_0) = \prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1}) \quad (5)$$

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{1 - \beta_t}\mathbf{x}_{t-1}, \beta_t \mathbf{I}) \quad (6)$$

and the reverse process:

$$p(\mathbf{x}_{0:T}) = p(\mathbf{x}_T) \prod_{t=1}^T p(\mathbf{x}_{t-1}|\mathbf{x}_t) \quad (7)$$

$$p(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mu(\mathbf{x}_t, t), \Sigma(\mathbf{x}_t, t)) \quad (8)$$

where $\mathbf{x}_i (i = 1, 2, 3, \dots, T)$ is the i^{th} input of the DM, t represents the time step or diffusion step in the diffusion process, β_t represents the variance schedule at the time step t , T represents the number of steps in the DM, p is the target distribution, q denotes a probability distribution, and $\mu(\mathbf{x}_t, t)$ and $\Sigma(\mathbf{x}_t, t)$ represent the mean and variance of the Gaussian distribution respectively.

In the conditional DM, although there exists the condition $\mathbf{y}$, the forward diffusion process is not affected. The forward process of the conditional DM is factorized as

$$\hat{q}(\mathbf{x}_{1:T}|\mathbf{x}_0, \mathbf{y}) = \prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1}) \quad (9)$$

where $\hat{q}$ denotes the probability distribution in the conditional DM after incorporating $\mathbf{y}$.

However, the condition $\mathbf{y}$ affects the progression of the reverse process by guiding the model to generate outputs that are consistent with the specified condition. The reversed process of the DM is tuned as

$$p(\mathbf{x}_{0:T}|\mathbf{y}) = p(\mathbf{x}_T) \prod_{t=1}^T p(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{y}) \quad (10)$$

B. Model Learning

1) *Overall Learning Strategy*: The learning process of the Causal VAE-DM is divided into training the VAE and training the conditional DM. When the confounders are observed, a classifier is incorporated into the loss function to effectively address confounding factors. The overall loss function could be represented as

$$\mathcal{L} = \mathcal{L}_{\text{VAE}} + \mathcal{L}_{\text{DM}} + \mathcal{L}_{\text{CE}} \quad (11)$$

where $\mathcal{L}_{\text{VAE}}$ is the loss function of the VAE and the $\mathcal{L}_{\text{DM}}$ is the loss function of the DM while the $\mathcal{L}_{\text{CE}}$ is defined in (4).

2) *Learning Strategy of VAE*: The conventional VAE loss has two components: a reconstruction loss and a KL divergence term. The reconstruction loss, which uses MSE for continuous data or BCE for binary data, ensures the decoder accurately recovers the input, while the KL divergence regularizes the latent space. The reconstruction loss is factorized as

$$\mathcal{L}_{\text{reconstruction}} = -\mathbb{E}_{q(\mathbf{z}|\mathbf{x})} [\log p(\mathbf{x}|\mathbf{z})] \quad (12)$$

where $\mathbb{E}_Y[f(Y)]$ represents the function $f(Y)$, which is averaged over all possible values weighted by the probability.

KL divergence encourages the latent distribution $q(\mathbf{z}|\mathbf{x})$ to be close to a prior distribution, typically a standard Gaussian $p(\mathbf{z}) = \mathcal{N}(0, \mathbf{I})$. The KL divergence ensures that the learned latent space is smooth and enables the generation of new samples by sampling from the prior distribution. The loss function of the KL divergence is

$$\mathcal{L}_{\text{KL}} = \int q(\mathbf{z}|\mathbf{x}) \log \frac{q(\mathbf{z}|\mathbf{x})}{p(\mathbf{z})} d\mathbf{z} \quad (13)$$

To ensure the identifiability of the adjacency matrix $\mathbf{A}$, the learning guidance $\mathbf{u}$ is used to guide the identifiability. $\mathbf{u}$ can be leveled by minimizing the loss function l_u :

$$l_u = \mathbb{E}_X \|\mathbf{u} - \sigma(\mathbf{A}^T \mathbf{u})\|_2^2 \quad (14)$$

where $\mathbf{u}$ is the label, σ is a logistic function and the notation $\|\cdot\|_2^2$ represents the squared L_2 -norm of a vector.

In a similar method, $\mathbf{z}$ is regulated by the loss function l_z :

$$l_z = \mathbb{E}_{\mathbf{z} \sim q_\phi} \left[\sum_{i=1}^n \|z_i - g_i(\mathbf{A}_i \circ \mathbf{z}; \eta_i)\|_2^2 \right] \quad (15)$$

where n is the number of the causal factors, g_i is the i^{th} element of a set of mild nonlinear and invertible functions $\mathbf{g}$, the symbol $\circ$ denotes the elementwise product and η_i is the parameter of the function set $\mathbf{g}$.

The loss function of the VAE in the Causal VAE-DM is

$$\mathcal{L}_{\text{VAE}} = \mathcal{L}_{\text{reconstruction}} + \mathcal{L}_{\text{KL}} + \gamma l_u + \lambda l_z \quad (16)$$

where the γ and λ are the coefficients of the constraints l_u and l_z .

3) *Learning Strategy of DM*: For the conditional DM, the label $\mathbf{u}$ is also used to guide the learning process. Given an image $\mathbf{x}_0$, the loss function is factorized with the condition $\mathbf{y}$:

$$\mathcal{L}_{\text{DM}} = \mathbb{E}_{\mathbf{z} \sim q_{\psi}(\mathbf{z}|\mathbf{y}, \mathbf{x}_0)} \left[\mathbb{E}_{q_{\phi}(\mathbf{x}_{1:T}|\mathbf{y}, \mathbf{z}, \mathbf{u}, \mathbf{x}_0)} \log \frac{q_{\phi}(\mathbf{x}_{1:T} | \mathbf{y}, \mathbf{z}, \mathbf{u}, \mathbf{x}_0)}{p_{\psi}(\mathbf{x}_{0:T} | \mathbf{y}, \mathbf{z}, \mathbf{u})} \right] \quad (17)$$

C. Model Intervention

With the loss function (11), the Causal VAE-DM could obtain the desired causal latent. The intervention is deployed during the testing to evaluate the performance of the Causal VAE-DM. The intervention process refers to actively changing or setting the value of a variable (or variables) to observe how this manipulation affects other variables in the system. This process is foundational for distinguishing causation from correlation, as it allows researchers to observe the direct effects of changes in one variable on others. The intervention process is commonly denoted by *do*-calculus, where an intervention is represented as $do(X = x)$. This operation means that the variable X is set to the value x , irrespective of its natural causes.



(a) Original Images from the MNIST Dataset



(b) $do(\text{Thickness} = 2)$ from Causal VAE-DM



(c) $do(\text{Thickness} = 4)$ from Causal VAE-DM

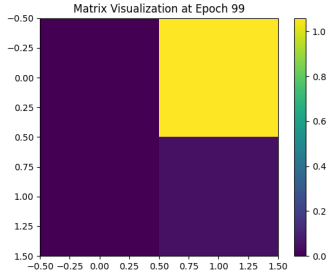


(d) $do(\text{Thickness} = 2)$ from Conventional CausalVAE

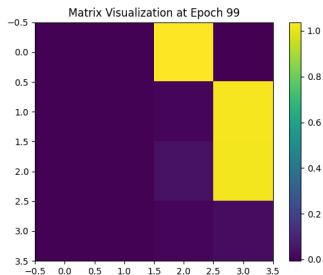


(e) $do(\text{Thickness} = 4)$ from Conventional CausalVAE

Fig. 3: Results on the MNIST Dataset using two different methods



(a) The causal matrix A of the MNIST Dataset



(b) The causal matrix A of the Flow Dataset

Fig. 2: Causal matrix A at the 100th epoch

IV. EXPERIMENT

In this section, the proposed Causal VAE-DM model is evaluated through comprehensive experiments conducted on two datasets. The evaluations are designed to demonstrate the model's interpretability, effectiveness in learning causal relationships and capability to generate accurate and trustworthy counterfactual images. The experiments are carried out on a high-performance server configured with Ubuntu 20.04 and equipped with two NVIDIA RTX A6000 GPUs. The details are elaborated as follows, including the datasets, network design, and experimental results, with a focus on the improvements achieved by the Causal VAE-DM compared to conventional methods.

A. Datasets

To evaluate the interpretability of the Causal VAE-DM, the model has been trained by two datasets:

MNIST: The MNIST dataset [21] comprises 70,000 grayscale images of handwritten digits (0–9) at 28×28 pixels, divided into 51,000 for training, 9,000 for validation and 10,000 for test images, serving as a standard benchmark for image recognition and deep learning.

Flow Dataset: The Flow Dataset [18] contains 7,000 RGBA images (96×96 pixels) illustrating a ball interacting with fluid

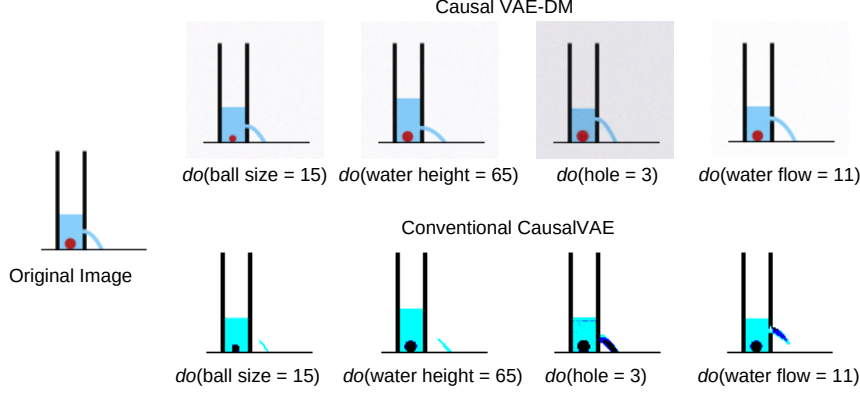


Fig. 4: Results on the Flow Dataset using two different methods

in a broken vessel, split into 5,100 for training, 900 for validation and 1,000 for testing images. In this dataset, the ball size influences water height, which affects water flow, while the hole size also impacts the water flow.

B. Experiment Details

The Causal VAE-DM is trained on each dataset individually. Due to the different types of images, the VAE part and the DM part are also different for each dataset.

In the MNIST dataset, it is found that the digit itself is the confounder in the causal relationship. It implies that when the cause is changed, the digit is also changed. To address the confounder, the classifier is introduced into the Causal VAE-DM. Furthermore, no observed confounder is found in the Flow dataset so there is no classifier in the model.

C. Experimental Results

1) *Causal graphs*: In Fig. 2, the images appear to be a heatmap visualization of a matrix at the 100th epoch. In Fig. 2(a), on both axes, the section from -0.50 to 0.50 indicates the cause. At the same time, the section from 0.50 to 1.50 indicates the effect. The yellow part illustrates that the cause could influence the effect, which means the factor thickness could influence the intensity. It is similar to Fig. 2(b). On both axes, the section from -0.50 to 0.50 is the factor ball size. The section from 0.50 to 1.50 is the factor hole. Moreover, the section from 1.50 to 2.50 is the factor water height. The section from 2.50 to 3.50 is the factor water flow. The results in Fig. 2(a) and Fig. 2(b), which means the model learns correct causal relationships.

2) *Results of the MNIST Dataset*: To evaluate the results of the intervention, the experiments are deployed on two datasets. For the MNIST, the range of the label thickness is from 1 to 5.8 and the range of the intensity is from 67 to 255. The results of the intervention are shown in Fig. 3. The images in Fig. 3(a) are the original images from the dataset. The images in Fig. 3(b) and Fig. 3(c) are the results yielded by the Causal VAE-DM when $do(\text{Thickness} = 2)$ and $do(\text{Thickness}$

$= 4)$, respectively. To evaluate the performance of the Causal VAE-DM. The images in Fig. 3(d) and Fig. 5(e) are the results from the conventional CausalVAE with the same intervention. The ideal intervention occurs when a decrease in thickness leads to a corresponding decrease in intensity and an increase in thickness results in a proportional increase in intensity. The results of the Causal VAE-DM correspond to the ideal intervention. However, for the conventional CausalVAE, the change of intensity is not clear. Moreover, when the thickness is higher, the images show some errors in the background, which do not appear in the Causal VAE-DM.

3) *Results of the Flow Dataset*: The results on the Flow Dataset are shown in Fig. 4. In the Flow Dataset, the range of the label ball size is from 5 to 35. The range of the hole is from 0 to 4. The range of the water height is from 33 to 85 and the range of the water flow is from 6 to 15. From the intervention results, the Causal VAE-DM and the conventional CausalVAE can both learn the causal relationships. When the ball size is changed, the water height is also changed. With the water height changes, the water flow is influenced. The result proves the interpretability and trustworthiness of the model. Furthermore, compared with the conventional CausalVAE, the results from the Causal VAE-DM show better details in the images.

TABLE I: MAE Comparison on MNIST and Flow Datasets

Dataset	Intervention	Causal VAE-DM	Conventional CausalVAE
MNIST	$do(\text{thickness} = 2)$	0.06	1.01
	$do(\text{thickness} = 4)$	0.081	0.845
	$do(\text{ball size} = 15)$	0.023	16.6553
Flow	$do(\text{water height} = 65)$	0.021	19.1042
	$do(\text{hole} = 3)$	0.091	8.7898
	$do(\text{water flow} = 11)$	0.011	9.7236

4) *Results Analysis*: To evaluate the accuracy of the model's control over latent factors, the Mean Absolute Error (MAE) is used. TABLE I illustrates the comparison of the MAE between the Causal VAE-DM and Conventional CausalVAE. The blue part is the MAE of the Causal VAE-DM while

the orange part is the MAE of the Conventional CausalVAE. As indicated in TABLE I, the MAE of the Causal VAE-DM is much lower than that of the Conventional CausalVAE, ascertaining the competence of the Causal VAE-DM against Conventional CausalVAE for photo-realistic image generation.

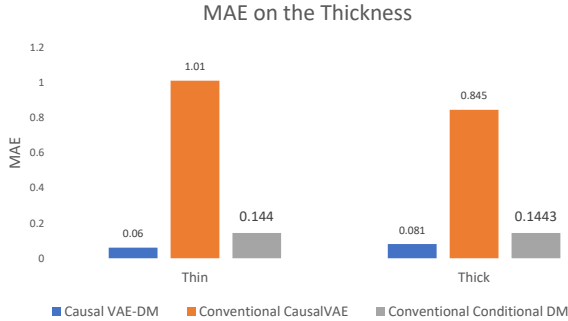


Fig. 5: MAE on the Factor Thickness

Moreover, the models, like the Conventional Conditional DMs [22], are not able to change the effects by changing the causes. Only the performance of these models on the factor thickness is evaluated by MAE in Fig. 5 while the Causal VAE-DM shows a preferable performance in controlling the latent factors.

V. CONCLUSION

This study introduced Causal VAE-DM, a trustworthy architecture, to generate counterfactual images. The model effectively disentangles and manipulates latent factors, as evidenced by lower MAE scores and superior image quality compared to conventional methods. The experimental results underscore its value in advancing the development of interpretable and reliable generative models. Additionally, with the results on different causal graphs, the Causal VAE-DM demonstrates its adaptability and consistent performance across various environments. In future work, the Causal VAE-DM will be enhanced to identify and disentangle a broader range of causal factors. Furthermore, its application will be extended to engineering domains, providing innovative solutions to practical problems and advancing the integration of causal learning in real-world systems. By enabling precise interventions and generating counterfactual images aligned with real-world causal relationships, the Causal VAE-DM ensures trustworthiness.

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